Seong-Jun Kang, Ph.D.

Computational immunologist working in single-cell and spatial multi-omics of the immune system

Research Scientist applicant · Hyman Lab, MassGeneral Institute for Neurodegenerative Disease · transcriptomics and neurodegeneration

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PROFESSIONAL STATUS

I'm an immunologist working in computational single-cell analysis. I read immune cell state from large, incomplete single-cell and spatial data, and I check that it holds on held-out data. The brain work I have done is single-cell immune profiling in an Alzheimer's mouse model, where I built the path from tissue harvest through 10× to the analysis.

RESEARCH OBJECTIVE

to add an immune-cell layer to the transcriptomic dissection of the human Alzheimer's brain, and to separate immune programs that drive neurodegeneration from those that only track it. I write my own funding: lead proposal writer on ~$1.3M of funded industry grants and a >30% contributor to the ~$1.6M SCAID atlas proposal.

SCIENTIFIC INTERESTS

single-cell and spatial multi-omics; immune cell-state inference from incomplete data; consensus NMF and isoform-resolved analysis; neuro-immune interactions in neurodegeneration; atlas-scale QC across many donors and sites; calibrated, held-out-validated ML.

WHAT I WOULD BRING

- **Brain immune single-cell analysis.** In the 5×FAD Alzheimer's mouse model I built the analysis path from end to end, running whole-brain harvest, enzymatic digestion, Percoll enrichment, and FACS into 10× and then the analysis. It resolved ten immune populations from 11,587 cells and showed exhausted CD8 tissue-resident memory T cells engaging the choroid plexus epithelium at the blood-CSF barrier, while that epithelium turned on MHC class I, interferon-stimulated genes, and JAK-STAT signaling (Journal of Neuroimmunology 2025, first author).
- **Cell-state calling and held-out testing.** From 251,055 memory CD4 T cells with paired T-cell receptors I used consensus NMF to find twelve reproducible gene programs. Five-prime single-cell reads cannot separate the short and long PDE4D isoforms, so I wrote a UMI-aware pipeline that goes back to the aligned reads and assigns the isoform per cell. The short-isoform state recurs across 702 samples and 23 conditions in the SCAID atlas, and a classifier on subset composition reaches a held-out AUC of 0.78. The integration, the NMF, the isoform pipeline, and the classifier are my work, and colleagues carried out the wet validation (co-first author, in revision at Nature Immunology).
- **Methods that transfer.** Olink targeted plasma proteomics, GeoMx region-of-interest microdissection and profiling, and high-parameter flow panels are assays I have run, and I designed the causal overexpression test in the PDE4D study. I ran them on immune cells and tumor tissue, and I have not worked with human brain tissue. What transfers is the method rather than the biology, and the biology is what I would be learning from the people who already do it.
- **A first six months.** In the first three months I would take one human brain single-nucleus dataset the group already has and annotate the immune and perivascular compartment, writing the marker logic down and reporting uncertainty per call, with no new wet work. In the next three months I would test the calls on held-out donors, run consensus NMF for the programs, and check whether program scores track pathology stage rather than tissue batch or postmortem interval. The go/no-go at six months is whether the calls recover states already known from sorted-cell references and whether they hold out of sample.
- **Grants written.** In industry I was the lead proposal writer on two Korea Drug Development Fund awards for an anti-CD40 antibody (about $571,000 and $321,000) and on two Ministry of SMEs and Startups awards, and I did the proposal and study design for a NAVER-funded engineered Treg program. Those come to about $1.3 million of funded work. During my doctorate I wrote more than 30 percent of the proposal for SCAID, the Korean single-cell atlas of immune disease, funded at about $1.6 million, and I then ran its multi-site QC, milestone tracking and multi-PI coordination. All of these were held by the institutional principal investigator, so my role was proposal writing and project execution, not as PI.

EDUCATION

DEGREES

Ph.D. 2014–2023
Seoul National University, Seoul, Republic of Korea
Department of Biomedical Sciences
Thesis: Exploration of the Role of Regulatory T cells in Various Pathogenic Contexts
Advisor: Chung-Gyu Park, M.D., Ph.D.

M.S. 2012–2014
Seoul National University, Seoul, Republic of Korea
Department of Biomedical Sciences
Thesis: Enhanced LPS-Induced Monocyte Activation by Recombinant Human Soluble CD14
Advisor: Chung-Gyu Park, M.D., Ph.D.

B.S. 2008–2012
Sungkyunkwan University, Suwon, Republic of Korea
Department of Genetic Engineering

RESEARCH & PROFESSIONAL EXPERIENCE

Researcher (postdoctoral) 2025–present
Yonsei University Wonju College of Medicine · Dept. of Physiology / Organelle Medicine Research Center, Wonju, Republic of Korea
Deep-learning multimodal-integration models fusing imaging, transcriptomic, and clinical data into single predictive models.

Associate Director 2023–2025
PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea
Led a humanized anti-CD40 antibody program: candidate ranking by **surface plasmon resonance (Biacore)** and ELISA, receptor-blocking confirmation, human and non-human-primate cross-reactivity, in vivo efficacy in primates, and the regulatory documentation. Placed and oversaw CRO and core-facility work, from study design and assay conditions through data delivery and go/no-go calls. Built the in-house single-cell and spatial analysis pipeline. Ran cynomolgus and marmoset EAE studies for multiple sclerosis programs.

Manager 2021–2023
PB Immune Therapeutics Inc. · Dept. of Basic Research, Seoul, Republic of Korea
Wrote the two KDDF proposals that funded the anti-CD40 program and ran the lead-selection work under them.

Research Student 2020–2023
SCAID, Single Cell Atlas of Immune Diseases, Genomic Medicine Institute, Seoul National University, Seoul, Republic of Korea
Contributed the founding proposal; executed within the funded program (700+ donors × 8 organs): milestone tracking, multi-site QC, multi-PI coordination.

Researcher 2019–2021
Xenotransplantation Research Center, Seoul National University, Seoul, Republic of Korea
Rhesus macaque tissue processing and longitudinal immune monitoring across transplantation studies.

Teaching Assistant 2014–2016
Seoul National University College of Medicine · Dept. of Microbiology and Immunology, Seoul, Republic of Korea

TECHNICAL SKILLS

Brain and CNS tissue
Mouse brain immune profiling run end to end: whole-brain harvest, enzymatic digestion, Percoll enrichment, $CD45^+$ FACS, 10× library generation, and the analysis (*J Neuroimmunol* 2025, first author; 5×FAD model, 11,587 cells, 10 immune populations). Neuro-immune analysis of choroid plexus epithelium and infiltrating $CD8^+$ resident-memory T cells at the blood–CSF barrier. Non-human-primate CNS tissue, brain and spinal cord, across rhesus, cynomolgus and marmoset studies, including EAE models of multiple sclerosis.

Cell-state calling and its checks	**Consensus NMF** gene programs (k = 14 → 12 reproducible programs across 251,055 memory $CD4^+$ T cells). Custom UMI-aware, BAM-level isoform code (Python) resolving PDE4D short- versus long-isoform identity from 5' reads, which a count matrix cannot separate. **Milo** KNN-graph differential abundance; **scVI/scANVI** integration; condition classifiers reported with held-out AUC and confidence intervals. Multi-site sample QC, milestone tracking and multi-PI coordination across a 700-donor, 8-organ atlas (SCAID).
Binding & protein interaction	**Surface plasmon resonance (Biacore)** run hands-on for antibody lead work: association and dissociation rates, affinity ranking across a candidate panel, receptor-blocking confirmation, and human / non-human-primate cross-reactivity. **ELISA**-based affinity and epitope analysis; competition binding. Antibody lead selection and optimization from candidate panel to registered patent. Functional validation of a peptide inhibitor of calcineurin in primary human T cells (CABIN1 core peptide, *Exp Mol Med* 2022).
Program & external execution	CRO and core-facility work from study design and placement through data delivery and go/no-go calls; assay conditions specified to the vendor; regulatory documentation for a therapeutic antibody program; funder milestones and reporting; multi-site sample QC across a 700-donor collection.
Immunological assays	High-dimensional flow cytometry (panels up to 24 markers); multiplex panel design; 10-color 6-way FACS. **Olink** targeted plasma proteomics (immune & neurology panels). Primary human T-cell isolation, expansion and functional assays; monocyte and dendritic-cell differentiation; mesenchymal stem cell culture. Enzymatic tissue digestion, Percoll enrichment, cell and nucleus isolation.
Single-cell / spatial / multi-omic	scRNA-seq, scTCR-seq, snRNA/snATAC; 10× 5'/3' library generation and **10× Chromium Connect** automated preparation. **Seurat, Scanpy, Squidpy, Monocle3, CellChat/NicheNet**; scVI/scANVI, batch integration, **consensus NMF**; Milo KNN-graph differential abundance, GLIPH, InferCNV. Custom UMI-aware, BAM-level isoform analysis code (Python). Spatial: **GeoMx DSP** run end to end, **Visium**, **MACSima** cyclic multiplexed immunofluorescence.
Computational	Python: scikit-learn, XGBoost/LightGBM/CatBoost, **PyTorch**; AUC-validated classifiers held out on independent cohorts; multimodal deep-learning integration (imaging + transcriptomic + clinical). Linux servers · R/RStudio · Git-based reproducible analysis.
LLM and vision-language models	**Finetuning.** LoRA adaptation of an open vision-language model (**Gemma 4 12B**, mlx-vlm) so that an amyloid pathology tile returns a structured JSON reading. The output schema is enforced during decoding, and the split is held out at the case level, checked on every run. Against the same model zero-shot on 800 held-out tiles, cored-plaque F1 went from 0.12 to 0.44 and cerebral amyloid angiopathy from 0.00 to 0.69. **Retrieval and agents.** A retrieval-augmented question-answering service written end to end: corpus chunking, BM25 and dense-vector retrieval fused by reciprocal rank, query expansion, and streamed answers that carry their sources. **FastAPI**, containerized, running on **Google Cloud Run**. It answers questions about this CV at kangseongjun.com.
Microscopy & animal models	Spinning-disk confocal (Nikon CSU-W1 SORA), point-scanning confocal (Leica SP8), two-photon; IHC/IF (frozen & FFPE) with quantitative image analysis. Non-human primates across three species (rhesus, cynomolgus, marmoset), including EAE and islet-transplantation studies; mouse immune profiling.

PUBLICATIONS

18	17	1	6
TOTAL PAPERS	PEER-REVIEWED	IN REVISION	FIRST / CO-FIRST

Google Scholar · Full Profile, Citations & h-index ↗

SELECTED PUBLICATIONS · FIRST / CO-FIRST AUTHOR

2026 Woogil Song*, **Seong-Jun Kang***, Taehee Stella Kim*, Seyoung Jung, Sung Hee Kim, Su Chan Park, Sung Hae Chang, Eun Young Lee, Tae-Gyun Kim, Hyun Je Kim, Jeong Seok Lee. **"Conserved PDE4D short-isoform bias defines pathogenic Th1.17 state across human inflammatory diseases."** *Nature Immunology*. IF 27.7 in revision

First author (co-first). ***Design*** *— four matched cohorts (healthy control, psoriasis, rheumatoid arthritis, ankylosing spondylitis), 16 donors each, n = 64; memory CD4⁺ T enrichment by negative selection, 4-plex pooling with SNP demultiplexing so donor and sequencing run are not confounded; 5′ scRNA-seq with paired TCR. 251,055 memory CD4⁺ T cells.* ***Analysis*** *— built custom UMI-aware, BAM-level code (Python) resolving PDE4D short- vs long-isoform identity from 5′ reads, which standard count matrices cannot separate; consensus NMF (k = 14 → 12 programs); scRNA + scTCR + snATAC + STAT3 ChIP-seq integration; Milo KNN-graph differential abundance; condition classifier on Th1.17 subset composition — arthritis AUC 0.780 (95% CI 0.665–0.896), healthy control 0.771, psoriasis 0.806.* ***Generalisation*** *— the PDE4D-high Th1.17 state was conserved across the SCAID atlas (702 samples, 23 immune diseases) and in independent SLE and arthritis synovial-fluid datasets.* ***Causal test*** *— short-isoform overexpression (PDE4D1/2/6) raised the IFN-γ⁺IL-17A⁺ fraction while the long isoform PDE4D5 did not; the PDE4D-selective inhibitor zatolmilast lowered it.*

2025 **Seong-Jun Kang***, Yong-Hee Kim*, Thuy Nguyen-Phuong*, Yijoon Kim, Jin-Mi Oh, Jae-chun Go, DaeSik Kim, Chung-Gyu Park, Hyunsu Lee, Hyun Je Kim. **"Immune cell-enriched single-cell RNA sequencing unveils the interplay between infiltrated CD8⁺ T resident memory cells and choroid plexus epithelial cells in Alzheimer's disease."** *Journal of Neuroimmunology*. IF 2.9 Download PDF

First author (co-first). Built the complete brain-immune single-cell analysis path (whole-brain harvest → enzymatic digestion → Percoll enrichment → FACS → 10× → analysis); identified exhausted CD8⁺ tissue-resident memory T cells engaging choroid-plexus epithelium (↑ MHC-I and interferon-stimulated genes) at the blood–CSF barrier. 11,587 cells, 10 immune populations.

2025 **Seong-Jun Kang***, Jeong-Ryeol Gong*, Seon-Pil Jin, Jin-Mi Oh, Hyunjin Jin, Yuji Lee, Yewon Moon, Dongjun Kim, Hyo Jeong Nam, Hyun Seung Choi, Sanha Hwang, Yun Jung Huh, Kyung Yeon Han, Jihwan Moon, Jongsuk Chung, Woong-Yang Park, Chung-Gyu Park, Hyun Je Kim, Jeong Eun Kim. **"Deciphering Dysfunctional Regulatory T Cells in Atopic Dermatitis."** *Allergy*. IF 12.6 Download PDF

First author (co-first). Single-cell and functional dissection of dysfunctional skin-homing (CLA⁺) regulatory T cells in atopic dermatitis; OX40 (TNFRSF4) up-regulated alongside impaired suppressive function.

2024 Jong-Min Kim*, **Seong-Jun Kang***, So-Hee Hong*, Hyunwoo Chung, Jun-Seop Shin, Byoung-Hoon Min, Hyun Je Kim, Jongwon Ha, Chung-Gyu Park. **"Long-term control of diabetes by tofacitinib-based immunosuppressive regimen after allo-islet transplantation in diabetic rhesus monkeys that rejected previously transplanted porcine islets."** *Xenotransplantation*. IF 3.9 Download PDF

First author (co-first). Long-term diabetes control with a tofacitinib-based regimen after allo-islet transplantation in rhesus macaques; longitudinal immune monitoring.

2022 Youngkyoung Lim*, **Seong-Jun Kang***, Beom Keun Cho, Hyun Je Kim, … Chong Hyun Won, Chung-Gyu Park. **"Intra-tumoral heterogeneity and immune escape of melanoma arising from congenital melanocytic nevus revealed by spatial gene expression profiling."** *JEADV*. IF 9.2 Download PDF

First author (co-first). Ran the GeoMx DSP workflow end to end — assay setup and instrument operation, immunofluorescence staining (S100B / CD3ε / nuclei), fluorescence-guided ROI selection and imaging, barcode collection, then the image and spatial-transcriptomic analysis. ***Design*** *— one stage-IIC melanoma arising from a 4 cm congenital melanocytic nevus, segmented into four regions by invasion depth (M1–M4, 300 → 4,200 µm Breslow), four whole-transcriptome ROIs per region, 16 ROIs total, 18,677 genes. 2,067 differentially expressed genes (11.07%) at P < 0.0005; graded loss of HLA class I/II with depth, and B2M loss confirmed by immunohistochemistry.*

2019 Jong-Min Kim*, Jun-Seop Shin*, Byoung-Hoon Min*, **Seong-Jun Kang***, Il-Hee Yoon, Hyunwoo Chung, Jiyeon Kim, Eung-Soo Hwang, Jongwon Ha, Chung-Gyu Park. **"JAK3 inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus monkeys."** *Islets*. IF 2.2 Download PDF

Co-first author; one of four authors marked as contributing equally. JAK3-inhibitor-based immunosuppression in allogeneic islet transplantation in cynomolgus macaques.

CONTRIBUTING AUTHOR

2026 Jung Ho Lee*, Sung Ha Lim*, Hyungdon Kook, **Seong-Jun Kang**, Geon Lee, Brian Hyohyoung Lee, Hyunsung Nam, YongJun Kim, Christine Suh-Yun Joh, Soyoung Jeong, Dongryeol Shin, Hyun Je Kim, Jiyoung Ahn. "**Mycosis Fungoides-like Atopic Dermatitis Represents a Th22-dominant Inflammatory Endotype.**" ***Allergy***. IF 12.6

Performed the scRNA-seq & scTCR-seq analysis (scVI cross-dataset integration with public MF/AD cohorts, InferCNV genomic-identity inference, scTCR clonality). Showed mfAD is a reactive, oligoclonal Th22 endotype rather than a malignancy, responsive to JAK1 inhibition. 248,881 cells.

2024 Brian Hyohyoung Lee*, Yoon Ji Bang*, Sung Ha Lim*, **Seong-Jun Kang**, Sung Hee Kim, Seunghee Kim-Schulze, Chung-Gyu Park, Hyun Je Kim, Tae-Gyun Kim. "**High-dimensional profiling of regulatory T cells in psoriasis reveals an impaired skin-trafficking property.**" ***eBioMedicine***. IF 11.1 Download PDF

Co-author. High-dimensional immune profiling of regulatory T cells in psoriasis, revealing an impaired skin-trafficking phenotype.

2024 Youngkyoung Lim*, Beom Keun Cho*, **Seong-Jun Kang**, Soyoung Jeong, Hyun Je Kim, Jiyoon Baek, Ji Hwan Moon, Cheol Lee, Chan-Sik Park, Je-Ho Mun, Chong Hyun Won, Chung-Gyu Park. "**Spatial transcriptomic analysis of tumour-immune cell interactions in melanoma arising from congenital melanocytic nevus.**" ***JEADV***. IF 9.2 Download PDF

Co-author. Spatial transcriptomic (GeoMx) analysis of tumour-immune cell interactions in nevus-arising melanoma.

2023 Sunyoung Jung*, Sunho Lee, Hyun Je Kim, Sueon Kim, Ji Hwan Moon, Hyunwoo Chung, **Seong-Jun Kang**, Chung-Gyu Park. "**Mesenchymal stem cell-derived extracellular vesicles subvert Th17 cells by destabilizing RORγt through posttranslational modification.**" ***Experimental & Molecular Medicine***. IF 12.8

Co-author. MSC-derived extracellular vesicles suppress Th17 cells by destabilizing RORγt via post-translational modification.

2022 Sangho Lee, Han-Teo Lee, Young Ah Kim, Il-Hwan Lee, **Seong-Jun Kang**, Kyeongpyo Sim, Chung-Gyu Park, Kyungho Choi, Hong-Duk Youn. "**The optimized core peptide derived from CABIN1 efficiently inhibits calcineurin-mediated T-cell activation.**" ***Experimental & Molecular Medicine***. IF 12.8

Co-author. A CABIN1-derived core peptide blocks the calcineurin-CABIN1 interaction and inhibits calcineurin-mediated T-cell activation. My part was the T-cell functional validation.

2021 So Hee Hong, Hyun Je Kim, **Seong-Jun Kang**, Chung-Gyu Park. "**Novel Immunomodulatory Approaches for Porcine Islet Xenotransplantation.**" ***Current Diabetes Reports***. IF 4.2 Download PDF

Co-author. Review of novel immunomodulatory strategies for porcine islet xenotransplantation.

2020 Sunho Lee, Sueon Kim, Hyunwoo Chung, Ji Hwan Moon, **Seong-Jun Kang**, Chung-Gyu Park. "**Mesenchymal stem cell-derived exosomes suppress proliferation of T cells by inducing cell cycle arrest through p27kip1/Cdk2 signaling.**" ***Immunology Letters***. IF 4.4

Co-author. MSC-derived exosomes suppress T-cell proliferation through p27kip1/Cdk2-mediated cell-cycle arrest.

2019 Hyun Je Kim, Ji Hwan Moon, Hyunwoo Chung, Jun-Seop Shin, Bongi Kim, Jong Min Kim, Jung-Sik Kim, Il-Hee Yoon, Byoung-Hoon Min, **Seong-Jun Kang**, Yong-Hee Kim, Kyuri Jo, Joungmin Choi, Heejoon Chae, Won Woo Lee, Sun Kim, Chung-Gyu Park. "**Bioinformatic analysis of peripheral blood RNA-sequencing sensitively detects the cause of late graft loss following overt hyperglycemia in pig-to-non-human primate islet xenotransplantation.**" ***Scientific Reports***. IF 4.6

Co-author. Bioinformatic analysis of peripheral-blood RNA-seq to detect causes of late graft loss in pig-to-NHP islet xenotransplantation.

2018 Byoung-Hoon Min, Jun-Seop Shin, Jong-Min Kim, **Seong-Jun Kang**, Hyun-Je Kim, Il-Hee Yoon, Su-Kyoung Park, Ji-won Choi, Min-Suk Lee, Chung-Gyu Park. "**Delayed revascularization of islets after transplantation by IL-6 blockade in pig-to-non-human primate islet xenotransplantation model.**" ***Xenotransplantation***. IF 3.9

Co-author. IL-6 blockade delays islet revascularization in a pig-to-NHP islet xenotransplantation model.

2018 Jun-Seop Shin, Jong-Min Kim, Byoung-Hoon Min, Il Hee Yoon, Hyun Je Kim, Jung-Sik Kim, Yong-Hee Kim, **Seong-Jun Kang**, Jiyeon Kim, Hee-Jung Kang, Dong-Gyun Lim, Eung-Soo Hwang, Jongwon Ha, Sang-Joon Kim, Wan Beom Park, Chung-Gyu Park. **"Pre-clinical results in pig-to-non-human primate islet xenotransplantation using anti-CD40 antibody (2C10R4)-based immunosuppression."** *Xenotransplantation*. IF 3.9

Co-author. Pre-clinical pig-to-NHP islet xenotransplantation with anti-CD40 (2C10R4)-based immunosuppression.

2018 Jong-Min Kim, Jun-Seop Shin, Byoung-Hoon Min, Il Hee Yoon, **Seong-Jun Kang**, Won-Young Jeong, Sang-Joon Kim, Chung-Gyu Park. **"Pre-Clinical Results of Islet Allo-Transplantation Using JAK Inhibitor as Replacement for Tacrolimus in Cynomolgus Monkeys."** *Transplantation*. IF 6.2

Co-author. Pre-clinical islet allo-transplantation using a JAK inhibitor in place of tacrolimus in cynomolgus macaques.

2018 Jung-Sik Kim, Hyunwoo Chung, Nari Byun, **Seong-Jun Kang**, Sunho Lee, Jun-Seop Shin, Chung-Gyu Park. **"Construction of EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation."** *Biochemical and Biophysical Research Communications*. IF 3.1

Co-author. Engineered EMSC-islet co-localizing composites for xenogeneic porcine islet transplantation.

MANUSCRIPTS IN PREPARATION

2026 Kangsan Roh*, Sabyasachi Das*, **Seong-Jun Kang**, Olga Barreiro, Rodrigo Javier Gonzalez, Timothy P. Padera, Shraddha Shirguppe, Ahlim Lee, Youngju Song, Pazhanichamy Kalailingam, Claire E. Fong, Aarushi Gandhi, Meehyun Kim, Diana Tambala, Adam Vernon Crain, Casey A. Maguire, Ulrich H. Von Andrian, Christian L. Lino Cardenas, Sekeun Kim, Benjamin P. Kleinstiver, Mark E. Lindsay, Patricia L. Musolino, Rajeev Malhotra. **"In-vivo base editing of ACTA2 R179H restores lymphatic function and immune cell homeostasis in a murine model of Multisystemic Smooth Muscle Dysfunction Syndrome (MSMDS)."** in preparation

Co-author (immunology). Performed the plasma proteomics analysis (Olink immune & neurology panels, 12 patients vs 6 controls), showing broad down-regulation of lymphoid regulators (LTβR, BAFF) and leukocyte-adhesion molecules, with dimensionality reduction separating patients from controls; complemented by flow-cytometry immunophenotyping of lymph-node leukocytes (germinal-center B cells GL7⁺B220⁺).

RESEARCH GRANTS & FUNDING

INDUSTRY · PB IMMUNE THERAPEUTICS (MANAGER → ASSOCIATE DIRECTOR)

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Korea Drug Development Fund (KDDF)	Lead selection and optimization of a therapeutic anti-CD40 antibody for the treatment of multiple sclerosis	2021–2023	**~$571,000**	*Lead proposal writer (>90% of proposal & execution)*
Korea Drug Development Fund (KDDF)	Evaluation of drug efficacy and discovery of pharmacodynamic biomarkers of a new anti-CD40 antibody lead compound (PB101) for the treatment of pemphigus vulgaris	2022–2024	**~$321,000**	*Lead proposal writer (>90% of proposal & execution)*
Ministry of SMEs and Startups (MSS)	Scale-up Funding for Tech Commercialization	2022–2023	**~$222,000**	*Lead proposal writer; managed >90% of execution*
Ministry of SMEs and Startups (MSS)	Early-stage startup R&D: anti-CD40 biologic (PB-40N) to prevent anti-drug antibodies	2021–2022	**~$74,000**	*Lead proposal writer & core researcher*
NAVER Corporation	Fundamental therapeutic strategy for immune-mediated diseases using engineered Tregs (PB20X_Treg)	2023–2026	**~$128,000**	*Proposal & study design (~50%)*

DOCTORAL TRAINING · SEOUL NATIONAL UNIVERSITY

FUNDING AGENCY / SPONSOR	PROJECT	PERIOD	BUDGET (USD)	ROLE / CONTRIBUTION
Ministry of Science and ICT (MSIT) · Bio & Medical Technology Development Program	Establishing a meta-platform for discovering molecular targets by building a multi-layered immune-disease single-cell map (A9-22-1-05; SCAID, Single-cell Atlas for Immunological Disease)	2022–2028 (3+3)	**~$1.6M**	*Participated in proposal writing (>30%)*
Ministry of Education (MOE) & Ministry of Science and ICT (MSIT)	Laboratory-Specialized Tech-Transfer & R&BD Grant	2020–2021	**~$40,000**	*Grant writing, experimental design & technical validation*

All grants were held by the institutional Principal Investigator (Ph.D. advisor at SNU; company CEO/CSO at PB Immune Therapeutics); the listed roles reflect the applicant's contribution to proposal writing and project execution, ***not as PI****. Budgets converted at ≈ ₩1,400 / USD.*

RESEARCH INTERESTS

- **Reading immune cell state from incomplete single-cell data.** Cell-state calling under ambiguity, isoform and transcript resolution from short reads, and embedding representations of cells, with calibrated confidence.
- **Multi-modal integration of molecular identity.** Transcriptome, chromatin, spatial, and proteome into cell-state-resolved models.
- **Neuro-immune interactions in neurodegeneration.** T cells, microglia, and the blood-CSF barrier in Alzheimer's and related disease; ligand-receptor cross-talk and clonal/repertoire structure.
- **Building and checking computational analysis paths.** Custom callers, reproducible end-to-end analysis code, and comparison against domain ground truth.
- **Atlas-scale single-cell analysis across diseases and species.** Large multi-donor atlases (SCAID), cross-disease meta-analysis, and non-human-primate-to-human translation.

AWARDS, PATENTS & SCHOLARSHIPS

PATENTS

2024 **Antagonistic Anti-CD40 Humanized Antibody.** Republic of Korea, Appl. No. 10-2024-0085972; registered 2024-07-01. Co-inventor.

2019 **Immunosuppression Composition Comprising JAK Inhibitor.** Republic of Korea, Appl. No. 10-2014-3490000; registered 2019-08-20.

AWARDS

2024 **Grand Prize**, 2024 GPTers AI Hackathon (Project GAIA Team)

2023 **Grand Prize**, SBA Smart Workathon (Work Process Innovation)

2017 **Best Poster Award**, American Transplant Congress, Chicago · **Best Presentation Award**, Korean Association of Immunologists

SCHOLARSHIPS

2014–2020 NRF Research Scholarship for Ph.D. Candidates · Dean's Lecture & Research Support Scholarship, SNU College of Medicine · BK21 Plus Research Scholarship

SELECTED PRESENTATIONS

Poster KAI Annual Meeting, Incheon (2022) · Cytokine, Hawaii (2022) · American Transplant Congress, Chicago (2017) · The Transplantation Society, Hong Kong (2016) · IPITA/IXA/CTS, Melbourne (2015) · TREG, Shanghai (2012)

REFERENCES

Chung-Gyu Park, M.D., Ph.D.
Professor, Microbiology & Immunology / Biomedical Sciences, SNU College of Medicine; CEO/CSO, PB Immune Therapeutics · chgpark@snu.ac.kr
Doctoral advisor and industry supervisor; directed the anti-CD40 antibody program.

Seung-Kuy Cha, Ph.D.
Professor, Dept. of Physiology & Global Medical Science, Yonsei University Wonju College of Medicine · skcha@yonsei.ac.kr
Most recent research supervisor (2025–2026).

Hyun Je Kim, M.D., Ph.D.
Assistant Professor, Microbiology & Immunology / Biomedical Sciences; Interdisciplinary Program in Artificial Intelligence (IPAI), SNU · tte9801@snu.ac.kr
Co-corresponding on the single-cell studies; supervised the computational work.

Jeong Seok Lee, M.D., Ph.D.
Associate Professor, Graduate School of Medical Science & Engineering, KAIST, Daejeon · chemami@kaist.ac.kr
Corresponding author, PDE4D / Nature Immunology study.